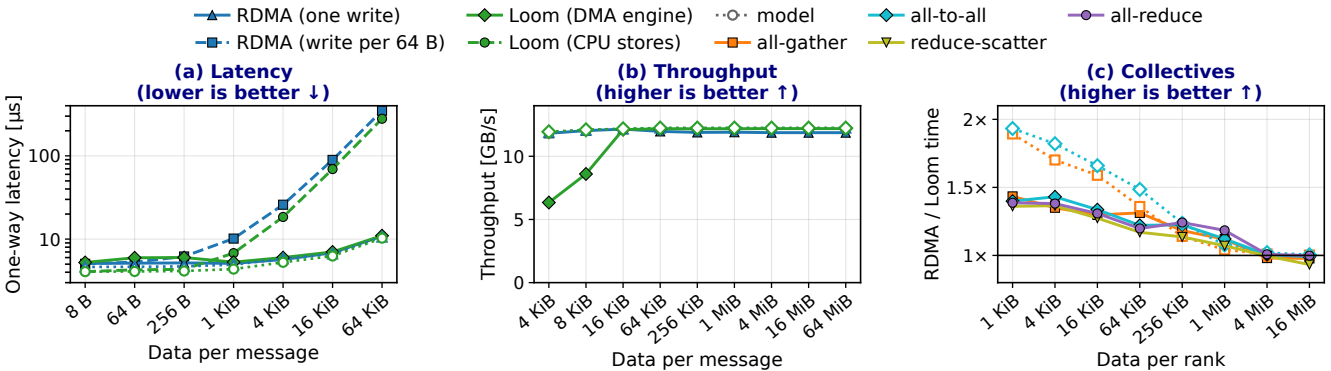
RDMA (one write)
RDMA (write per 64 B)
Loom (DMA engine)
Loom (CPU stores)
model
all-gather
all-to-all
reduce-scatter
all-reduce
(a) Latency
(lower is better ↓)
One-way latency [µs]
100
10
8 B
64 B
256 B
1 KiB
4 KiB
16 KiB
64 KiB
Data per message
(b) Throughput
(higher is better ↑)
Throughput [GB/s]
10
5
0
4 KiB
8 KiB
16 KiB
64 KiB
256 KiB
1 MiB
4 MiB
16 MiB
64 MiB
Data per message
(c) Collectives
(higher is better ↑)
RDMA / Loom time
2×
1.5×
1×
1 KiB
4 KiB
16 KiB
64 KiB
256 KiB
1 MiB
4 MiB
16 MiB
Data per rank